NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | SPL GTM-06 (AN) | Date: 19.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
C	A	D	B	C	A	C	D	A	B
11	12	13	14	15	16	17	18	19	20
D	C	D	D	B	D	D	C	D	B
21	22	23	24	25	26	27	28	29	30
45	1	50	144	7					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	D	C	C	A	C	C	B	B
36	37	38	39	40	41	42	43	44	45
D	A	B	D	A	A	C	B	A	C
46	47	48	49	50					
4	135	37	10	18					

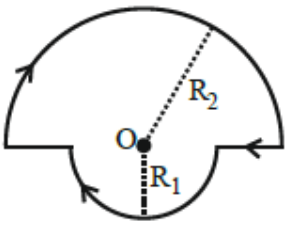
MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	D	B	B	C	B	B	A	A	D
61	62	63	64	65	66	67	68	69	70
A	D	C	D	C	A	B	D	C	D
71	72	73	74	75					
2	100	20	10	11					

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PHYSICS

1.



$$\frac{\mu_0 i}{2R_2}\left(\frac{\pi}{2\pi}\right)\otimes+\frac{\mu_0 i}{2R_1}\left(\frac{\pi}{2\pi}\right)\otimes$$

$$\left(\frac{\mu_0 i}{4R_2}+\frac{\mu_0 i}{4R_1}\right)\otimes$$

$$\frac{4\pi\times10^{-7}\times4}{4\times4\pi}+\frac{4\pi\times10^{-7}\times4}{4\times2\pi}$$

$$=3\times10^{-7}=\alpha\times10^{-7}$$

$$\alpha=3$$

2.

$$\phi_0=h\nu_0$$

$$6.63\times1.6\times10^{-19}=6.63\times10^{-34}\nu_0$$

$$\nu_0=\frac{1.6\times10^{-19}}{10^{-34}}$$

$$\nu_0=1.6\times10^{15}\text{Hz}$$

3.

$$\text{Potential}=\frac{kQ}{R}=\frac{k.Ze}{R}$$

$$=\frac{9\times10^9\times50\times1.6\times10^{-19}}{9\times10^{-13}\times10^{-2}}$$

$$=8\times10^6\text{V}$$

4.

$$\frac{V_1}{V_2}=\frac{N_1}{N_2}$$

$$\frac{230}{V_2}=\frac{10}{1}$$

$$V_2=23\text{V}$$

$$\text{Power consumed}=\frac{V_2^2}{R}$$

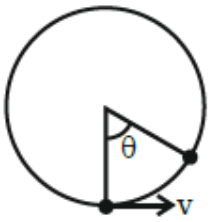
$$=\frac{23\times23}{46}=11.5\text{ W}$$

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5.	Co-efficient of friction depends on surface in contact So, depends on material of object.
6.	Let I_0 be intensity of unpolarised light incident on first polaroid. I_1 = Intensity of light transmitted from 1st polaroid $= \frac{I_0}{2}$ θ be the angle between 1st and 2nd polaroid ϕ be the angle between 2nd and 3rd polaroid $\theta + \phi = 90^0$ (as 1st and 3rd polaroid are crossed) $\phi = 90^0 - \theta$ I_2 = Intensity from 2nd polaroid $I_2 = I_1 \cos^2\theta = \frac{I_0}{2}\cos^2\theta$ I_3 = Intensity from 3rd polaroid $I_3 = I_2 \cos^2\phi$ $I_3 = I_1 \cos^2\theta\cos^2\phi$ $I_3 = \frac{I_0}{2}\cos^2\theta\cos^2\phi$ $\phi = 90 - \theta$ $I_3 = \frac{I_0}{2}\cos^2\theta\sin^2\theta$ $I_3 = \frac{I_0}{2}\left[\frac{2\sin\theta\cos\theta}{2}\right]^2$ $I_3 = \frac{I_0}{8}\sin^2 2\theta$ I_3 will be maximum when $\sin 2\theta = 1$ $2\theta = 90°$ $\theta = 45°$
7.	$i \propto \theta$ (angle of deflection) $\therefore \frac{i_2}{i_1} = \frac{\theta_2}{\theta_1} \Rightarrow \frac{i_2}{200\ \mu A} = \frac{\pi/10}{\pi/3} = \frac{3}{10}$ $\Rightarrow i_2 = 60\ \mu A$
8.	Kinetic energy $= \frac{f}{2}nRT$ $= \frac{5}{2}\times 1\times 8.31\times 300$ J $= 6232.5$ J

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9.



Loss in kinetic energy = Gain in potential energy

$$\Rightarrow \frac{1}{2}mv^2 = mg\ell(1-\cos\theta)$$

$$\Rightarrow \frac{v^2}{\ell} = 2g(1-\cos\theta)$$

Acceleration at lowest point = $\frac{v^2}{\ell}$

Acceleration at extreme point = $g\sin\theta$

Hence, $\frac{v^2}{\ell} = g\sin\theta$

$$\therefore \sin\theta = 2(1-\cos\theta)$$

$$\Rightarrow \tan\frac{\theta}{2} = \frac{1}{2} \Rightarrow \theta = 2\tan^{-1}\left(\frac{1}{2}\right)$$

10.

$${}_6C^{13} + \text{Energy} \rightarrow {}_6C^{12} + {}_0n^1$$

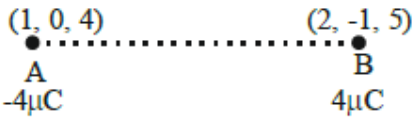
$\Delta m = (12.000000 + 1.008665) - 13.003354$

$= -0.00531$ u

$\therefore$ Energy required $= 0.00531 \times 931.5$ MeV

$= 4.95$ MeV

11.



$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\vec{p} = q\vec{\ell}$$

$$\vec{E} = 0.2\frac{V}{cm} = 20\frac{V}{m}$$

$$\vec{p} = 4\times\left(\hat{i}-\hat{j}+\hat{k}\right)$$

$$= \left(4\hat{i}-4\hat{j}+4\hat{k}\right)\mu C - m$$

$$\vec{\tau} = \left(4\hat{i}-4\hat{j}+4\hat{k}\right)\times\left(20\hat{i}\right)\times 10^{-6}\,Nm$$

$$= \left(8\hat{k}+8\hat{j}\right)\times 10^{-5} = 8\sqrt{2}\times 10^{-5}$$

$\alpha = 2$

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12.



$$f_c = \frac{v}{4\ell_1} \quad f_o = \frac{v}{2\ell_2}$$

$$|f_c - f_0| = 7$$

$$\frac{v}{4\times150} - \frac{v}{2\times350} = 7$$

$$\frac{v}{600\text{ cm}} - \frac{v}{700\text{ cm}} = 7$$

$$\frac{v}{6m} - \frac{v}{7m} = 7$$

$$v\left(\frac{1}{42}\right) = 7$$

$$v = 42\times7$$

$$=294\text{ m/s}$$

13.

$$P \propto T^3 \Rightarrow PT^{-3} = \text{constant}$$

$$PV^{\gamma} = \text{const}$$

$$P\left(\frac{nRT}{P}\right)^{\gamma} = \text{const}$$

$$P^{1-\gamma}T^{\gamma} = \text{const}$$

$$PT^{\frac{\gamma}{1-\gamma}} = \text{const}$$

$$\frac{\gamma}{1-\gamma} = -3$$

$$\gamma = -3 + 3\gamma$$

$$3 = 2\gamma$$

$$\gamma = \frac{3}{2}$$

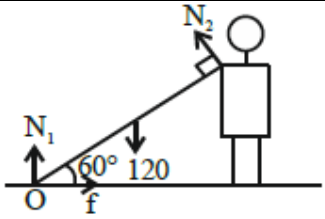
14.

$$a\sin\theta = \lambda \Rightarrow 0.001\sin\theta = 5\times10^{-7}$$

$$\Rightarrow \sin\theta = \frac{1}{2}$$

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15.



Torque about O = 0

$$120\left(\frac{L}{2}\cos 60°\right) - N_2 L = 0$$

$$N_2 = 30\ N$$

16.

$$[P] = \left[\frac{a}{V^2}\right] \Rightarrow [a] = [PV^2]$$

And [V] = [b]

$$\frac{[a]}{[b^2]} = \frac{[PV^2]}{[V^2]} = [P]$$

17.

$$\omega = \frac{2\pi}{T} \Rightarrow \omega \propto \frac{1}{T}$$

$T_{moon} = 27$ days

$T_{earth} = 365$ days 4 hour

$\Rightarrow \omega_{moon} > \omega_{earth}$

18.

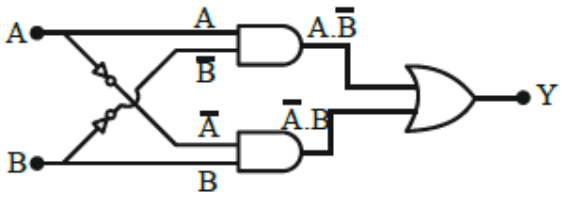
From KVL,

$V_1 + V_2 - V_3 = 0 \Rightarrow V_1 + V_2 = V_3$

19.

As specific resistance does not depends on dimension of wire so, it will not change.

20.

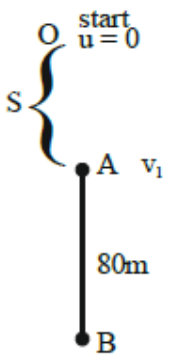


$Y = A \cdot \bar{B} + \bar{A} \cdot B$

This is XOR GATE

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21.



From $A \rightarrow B$

$$-80 = -v_1 t - \frac{1}{2} \times 10 t^2$$

$$-80 = -2v_1 - \frac{1}{2} \times 10 \times 2^2$$

$$-80 = -2v_1 - 20$$

$$-60 = -2v_1$$

$$v_1 = 30 \text{ m/s}$$

From O to A

$$v^2 = u^2 + 2gS$$

$$30^2 = 0 + 2 \times (-10)(-S)$$

$$900 = 20\ S$$

$$S = 45 \text{ m}$$

22.

$$X_c = \frac{1}{\omega C} = \frac{\pi}{2\pi \times 50 \times 10^{-3}} = 10\Omega$$

$$X_L = \omega L = 2\pi \times 50 \times \frac{100}{\pi} \times 10^{-3}$$

$$= 10\Omega$$

$\because X_C = X_L$,Hence, circuit is in resonance

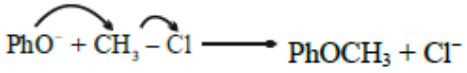
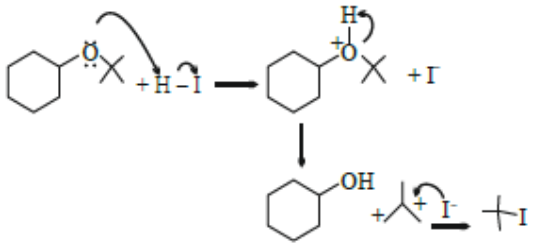
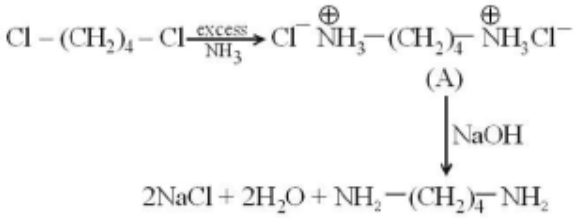
$$\therefore \text{ power factor } = \frac{R}{Z} = \frac{R}{R} = 1$$

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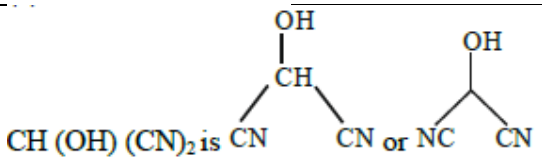
23.	Change in pressure $=\frac{1}{2}\rho v^2$ $4.5\times10^4-2.0\times10^4=\frac{1}{2}\times10^3\times v^2$ $2.5\times10^4=\frac{1}{2}\times10^3\times v^2$ $v^2=50$ $v=\sqrt{50}$ Velocity of water = $\sqrt{V}=\sqrt{50}$ $= V = 50$
24.	Longest wavelength corresponds to transition between n = 3 and n = 4 $\frac{1}{\lambda}=RZ^2\left(\frac{1}{3^2}-\frac{1}{4^2}\right)=RZ^2\left(\frac{1}{9}-\frac{1}{16}\right)$ $=\frac{7RZ^2}{9\times16}$ $\Rightarrow\lambda=\frac{144}{7R}$ for $Z=1$ $\quad\therefore\alpha=144$
25.	In pure rolling work done by friction is zero. Hence potential energy is converted into kinetic energy. Since initially the ring and the sphere have same potential energy, finally they will have same kinetic energy too. $\therefore$ Ratio of kinetic energies = 1 $\Rightarrow\frac{7}{x}=1\Rightarrow x=7$

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CHEMISTRY

26.	(A) → Kolbe Schmidt Reaction (B) → Reimer Tiemann Reaction (C) → Oxidation of phenol to p-benzoquinone (D) → $PhOH + NaOH \rightarrow H_2O + PhO^-$ 
27.	Boiling an egg causes denaturation of its protein resulting in loss of its quarternary, tertiary and secondary structures.
28.	In N^{3-} ion 'N' is present in its lowest possible oxidation state, hence it cannot be reduced further because of which it cannot act as an oxidizing agent.
29.	$HCOOH$, CH_3COOH ↑ Second homologue
30.	
31.	
32.	Carbylamine Test-Identification of primary amines Lucas Test - Differentiation between 1°, 2° and 3° alcohols Tollen's Test - Identification of Aldehydes Phthalein Dye Test - Identification of phenols
33.	$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution}}$ Since volume depends on temperature, molarity will change upon change in temperature.
34.	I > II > III, since neutral resonating structures are more stable than charged resonating structure. II > III, since stability of structure with –ve charge on more electronegative atom is higher.

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35.	Steam distillation is used for those liquids which are insoluble in water, containing non-volatile impurities and are steam volatile.
36.	$[Co(NH_3)_6]^{+3}$ – d^2sp^3 hybridization BrF_5 – sp^3d^2 hybridization $[PtCl_4]^{-2}$ – dsp^2 hybridization SF_6 – sp^3d^2 hybridization
37.	As tin coating is peeled off, then iron is exposed to atmosphere.
38.	Statement (1) is true, Ce^{+4} has noble gas electronic configuration. Statement (2) is also true due to high reduction potential for Ce^{4+}/Ce^{3+} (+1.74V), and stability of Ce^{3+}, Ce^{4+} acts as strong oxidizing agent.
39.	Statement-I: Oxygen can have oxidation state from –2 to +2, so statement I is incorrect Statement- II: On moving down the group stability of +4 oxidation state increases whereas stability of +6 oxidation state decreases down the group, according to inert pair effect. So both statements are wrong.
40.	$PhCH=CH_2 \xrightarrow{B_2H_6/H_2O_2OH^-} PhCH_2CH_2OH$ $PhCH_2CH_2OH + HBr \longrightarrow PhCH_2CH_2Br + H_2O\,(SN^{NGP})$ $PhCH_2CH_2CH_2OH \xleftarrow[(II)\ H_3O^+]{(I)\ H-\overset{\overset{O}{\|}}{C}-H} PhCH_2CH_2MgBr$ (Mg/dry ether)
41.	Eclipsed conformation is the least stable conformation of ethane.
42.	(1) Fluorspar is CaF_2
43.	 $CH(OH)(CN)_2$ is CN CN or NC CN
44.	$[Cr] = [Ar]4s^1\,3d^5$ $[Cd] = [Kr]5s^24d^{10}$ $[Cu] = [Ar]4s^13d^{10}$ $[Ag] = [Kr]5s^14d^{10}$ $[Zn] = [Ar]4s^23d^{10}$

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45. Since $CH_3-\overset{+}{C}H=CH$ is very unstable, CH_3–CH = CH – Cl cannot give $S_{N}1$ reaction.

46. The non-polar molecules are CO_2, H_2, CH_4 and BF_3

47.
$$\underset{(\text{Aniline MM}=93)}{C_6H_5NH_2} + CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3 \rightarrow$$

$$C_6H_5NH-\overset{O}{\overset{\|}{C}}-CH_3 + CH_3COOH$$
$$(\text{Acetanilide MM}=135)$$

$$n_{\text{Acetanilide}} = n_{\text{Aniline}}$$
$$\Rightarrow \frac{m}{135} = \frac{9.3}{93}$$
$$\Rightarrow m = 13.5\,g$$

48.
$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$
$$= 77.2\times10^3 - 400\times122 = 28400\,J$$
$$\Delta G^\circ = -2.303RT\log K$$
$$\Rightarrow 28400 = -2.303\times8.314\times400\log K$$
$$\Rightarrow \log K = -3.708 = -37.08\times10^{-1}$$

49.
$$\frac{t_{99.9\%}}{t_{1/2}} = \frac{\frac{2.303}{k}\left(\frac{a}{a-x}\right)}{\frac{2.303}{k}\log 2} = \frac{\log\left(\frac{100}{100-99.9}\right)}{\log 2} = \frac{\log 10^3}{\log 2} = \frac{3}{0.3} = 10$$

50.
$$2H^+_{(aq.)} + 2e^- \rightarrow H_2(g)$$
$$E_{cell} = E^0_{cell} - \frac{0.059}{2}\log\frac{P_{H_2}}{[H^+]^2}$$
$$= 0 - 0.059 \times 3 = -0.177 \text{ volts.} = -17.7 \times 10^{-2}\ V.$$

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51.

$2\tan^2\theta-5\sec\theta-1=0$

$\Rightarrow 2\sec^2\theta-5\sec\theta-3=0$

$\Rightarrow (2\sec\theta+1)(\sec\theta-3)=0$

$\Rightarrow \sec\theta=-\frac{1}{2},3$

$\Rightarrow \cos\theta=-2,\frac{1}{3}$

$\Rightarrow \cos\theta=\frac{1}{3}$

For 7 solutions n = 13

So, $\sum_{k=1}^{13}\frac{k}{2^k}=S$ (say)

$S=\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+....+\frac{13}{2^{13}}$

$\frac{1}{2}S=\frac{1}{2^2}+\frac{1}{2^3}+.....+\frac{12}{2^{13}}+\frac{13}{2^{14}}$

$\Rightarrow \frac{S}{2}=\frac{1}{2}.\frac{1-\frac{1}{2^{13}}}{1-\frac{1}{2}}-\frac{13}{2^{14}}\Rightarrow S=2.\left(\frac{2^{13}-1}{2^{13}}\right)-\frac{13}{2^{13}}$

52.

$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix}=0$$

$\Rightarrow (2\alpha+3)\left\{\frac{7\alpha}{6}\right\}-(3\alpha+1)\left\{\frac{-7}{6}\right\}=0$

$\Rightarrow (2\alpha+3).\frac{7\alpha}{6}+(3\alpha+1).\frac{7}{6}=0$

$\Rightarrow 2\alpha^2+3\alpha+3\alpha+1=0$

$\Rightarrow 2\alpha^2+6\alpha+1=0$

$\Rightarrow \alpha=\frac{-3+\sqrt{7}}{2},\frac{-3-\sqrt{7}}{2}$

Hence option (2) is correct.

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53.

$$I=\int_0^{\pi}\frac{dx}{1-2a\cos x+a^2};\ 0<a<1$$

$$I=\int_0^{\pi}\frac{dx}{1+2a\cos x+a^2}$$

$$2I=2\int_0^{\pi/2}\frac{2(1+a^2)}{(1+a^2)^2-4a^2\cos^2 x}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{2(1+a^2).\sec^2 x}{(1+a^2)^2.\sec^2 x-4a^2}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{2.(1+a^2).\sec^2 x}{(1+a^2)^2.\tan^2 x+(1-a^2)^2}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{\frac{2.\sec^2 x}{1+a^2}.dx}{\tan^2 x+\left(\frac{1-a^2}{1+a^2}\right)^2}$$

$$\Rightarrow I=\frac{2}{(1-a^2)}\left[\frac{\pi}{2}-0\right]$$

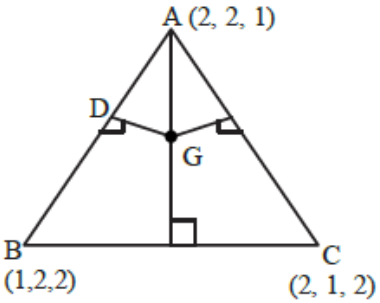
$$I=\frac{\pi}{1-a^2}$$

54.

ΔABC is equilateral

Orthocentre and centroid will be same

$$G\left(\frac{5}{3},\frac{5}{3},\frac{5}{3}\right)$$



Mid-point of AB is $D\left(\frac{3}{2},2,\frac{3}{2}\right)$

$$\therefore \ell_1=\sqrt{\frac{1}{36}+\frac{1}{9}+\frac{1}{36}}$$

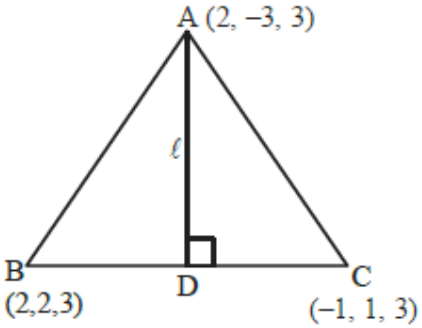
$$\ell_1=\sqrt{\frac{1}{6}}=\ell_2=\ell_3$$

$$\therefore \ell_1^2+\ell_2^2+\ell_3^2=\frac{1}{2}$$

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55. $AB = 5$

$AC = 5$



$\therefore$ D is midpoint of BC

$$D\left(\frac{1}{2},\frac{3}{2},3\right)$$

$$\therefore l=\sqrt{\left(2-\frac{1}{2}\right)^2+\left(-3-\frac{3}{2}\right)^2+(3-3)^2}$$

$$l=\sqrt{\frac{45}{2}}$$

$$\therefore 2l^2=45$$

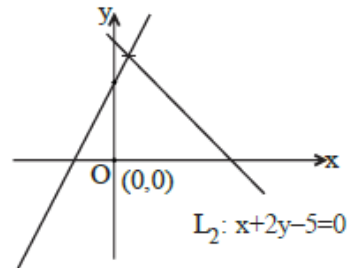
56. $P(a^2, a+1)$

$L_1 = 3x - y + 1 = 0$

Origin and P lies same side w.r.t. L_1

$\Rightarrow L_1(0) \cdot L_1(P) > 0$

$\therefore 3(a^2) - (a+1) + 1 > 0$



L_1: 3x–y+1=0

$\Rightarrow 3a^2 - a > 0$

$$a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

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Let $L_2 : x + 2y - 5 = 0$

Origin and P lies same side w.r.t. L_2

$\Rightarrow L_2(0).L_2(P) > 0$

$\Rightarrow a^2 + 2(a+1) - 5 < 0$

$\Rightarrow a^2 + 2a - 3 < 0$

$\Rightarrow (a+3)(a-1) < 0$

$\therefore a \in (-3,1)$ .....(2)

Intersection of (1) and (2)

$a \in (-3,0) \cup \left(\frac{1}{3},1\right)$

57.

$\tan^{-1}x + \tan^{-1}2x = \frac{\pi}{4}; x > 0$

$\Rightarrow \tan^{-1}2x = \frac{\pi}{4} - \tan^{-1}x$

Taking tan both sides

$\Rightarrow 2x = \frac{1-x}{1+x}$

$\Rightarrow 2x^2 + 3x - 1 = 0$

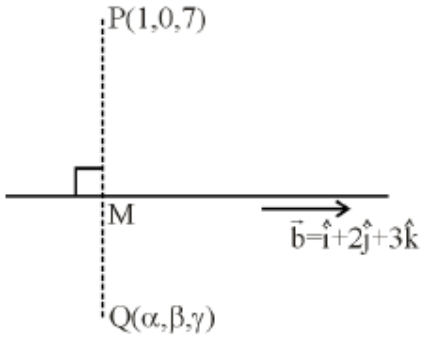
$x = \frac{-3 \pm \sqrt{9+8}}{8} = \frac{-3 \pm \sqrt{17}}{8}$

Only possible $x = \frac{-3+\sqrt{17}}{8}$

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58.

$$L_1 = \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \lambda$$



$M(\lambda, 1+2\lambda, 2+3\lambda)$

$\overrightarrow{PM} = (\lambda-1)\hat{i} + (1+2\lambda)\hat{j} + (3\lambda-5)\hat{k}$

$\overrightarrow{PM}$ is perpendicular to line L_1

$\overrightarrow{PM}\vec{b} = 0 \quad (\vec{b} = \hat{i}+2\hat{j}+3\hat{k})$

$\Rightarrow \lambda - 1 + 4\lambda + 2 + 9\lambda - 15 = 0$

$14\lambda = 14 \Rightarrow \lambda = 1$

$\therefore M = (1,3,5)$

$\vec{Q} = 2\vec{M} - \vec{P}$ [M is midpoint of $\vec{P}$ & $\vec{Q}$]

$\vec{Q} = 2\hat{i} + 6\hat{j} + 10\hat{k} - \hat{i} - 7\hat{k}$

$\vec{Q} = \hat{i} + 6\hat{j} + 3\hat{k}$

$\therefore (\alpha,\beta,\gamma) = (1,6,3)$

Required line having direction cosine (l, m, n)

$l^2 + m^2 + n^2 = 1$

$$\Rightarrow l^2 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 = 1$$

$$l^2 = \frac{1}{4}$$

$$\therefore l = \frac{1}{2}$$ [Line make acute angle with x-axis]

Equation of line passing through (1, 6, 3) will be

$$\vec{r} = (\hat{i}+6\hat{j}+3\hat{k}) + \mu\left(\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}\right)$$

Option (3) satisfying for $\mu = 4$

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59. $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots\ldots, -129\frac{1}{4}$

This is A.P. with common difference

$d_1 = -1 + \frac{1}{4} = -\frac{3}{4}$

$-129\frac{1}{4}, \ldots\ldots\ldots\ldots, 19\frac{1}{4}, 20$

This is also A.P. $a = -129\frac{1}{4}$ and $d = \frac{3}{4}$

Required term =

$-129\frac{1}{4} + (20-1)\left(\frac{3}{4}\right)$

$= -129 - \frac{1}{4} + 15 - \frac{3}{4} = -115$

60. $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0 \ \forall \ x \in (0, 3)$

$\Rightarrow f'(x)$ is increasing function

$g'(x) = 3 \times \frac{1}{3} f'\left(\frac{x}{3}\right) - f'(3-x)$

$= f'\left(\frac{x}{3}\right) - f'(3-x)$

If g is decreasing in $(0, \alpha)$

$g'(x) < 0$

$f'\left(\frac{x}{3}\right) - f'(3-x) < 0$

$f'\left(\frac{x}{3}\right) < f'(3-x)$

$\Rightarrow \frac{x}{3} < 3 - x$

$\Rightarrow x < \frac{9}{4}$

Therefore $\alpha = \frac{9}{4}$

Then $8\alpha = 8 \times \frac{9}{4} = 18$

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61.

$$\lim_{x\to 0}\frac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x}=\frac{1}{3}$$

$$\Rightarrow\lim_{x\to 0}\frac{3+\alpha\left[x-\frac{x^3}{3!}+....\right]+\beta\left[1-\frac{x^2}{2!}+\frac{x^4}{4!}....\right]+\left(-x-\frac{x^2}{2}-\frac{x^3}{3}...\right)}{3\tan^2 x}=\frac{1}{3}$$

$$\Rightarrow\lim_{x\to 0}\frac{(3+\beta)+(\alpha-1)x+\left(-\frac{1}{2}-\frac{\beta}{2}\right)x^2+....}{3x^2}\times\frac{x^2}{\tan^2 x}=\frac{1}{3}$$

$$\Rightarrow\beta+3=0,\ \alpha-1=0 \text{ and } \frac{-\frac{1}{2}-\frac{\beta}{2}}{3}=\frac{1}{3}$$

$$\Rightarrow\beta=-3,\alpha=1$$

$$\Rightarrow 2\alpha-\beta=2+3=5$$

62.

$$(x^2-4)dy-(y^2-3y)dx=0$$

$$\Rightarrow\int\frac{dy}{y^2-3y}=\int\frac{dx}{x^2-4}$$

$$\Rightarrow\frac{1}{3}\int\frac{y-(y-3)}{y(y-3)}dy=\int\frac{dx}{x^2-4}$$

$$\Rightarrow\frac{1}{3}(\ln|y-3|-\ln|y|)=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$$

$$\Rightarrow\frac{1}{3}\ln\left|\frac{y-3}{y}\right|=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$$

$$\text{At } x=4,\ y=\frac{3}{2}$$

$$\therefore C=\frac{1}{4}\ln 3$$

$$\therefore\frac{1}{3}\ln\left|\frac{y-3}{y}\right|=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+\frac{1}{4}\ln(3)$$

$$\text{At } x=10$$

$$\frac{1}{3}\ln\left|\frac{y-3}{y}\right|=\frac{1}{4}\ln\left|\frac{2}{3}\right|+\frac{1}{4}\ln(3)$$

$$\ln\left|\frac{y-3}{y}\right|=\ln 2^{3/4},\ \forall x>2,\frac{dy}{dx}<0$$

$$\text{as } y(4)=\frac{3}{2}\Rightarrow y\in(0,3)$$

$$-y+3=8^{1/4}.y$$

$$y=\frac{3}{1+8^{1/4}}$$

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63.

$x^2 - x - 1 = 0$

$S_n = 2023\alpha^n + 2024\beta^n$

$S_{n-1} + S_{n-2} = 2023\alpha^{n-1} + 2024\beta^{n-1} + 2023\alpha^{n-2} + 2024\beta^{n-2}$

$= 2023\alpha^{n-2}[1+\alpha] + 2024\beta^{n-2}[1+\beta]$

$= 2023\alpha^{n-2}[\alpha^2] + 2024\beta^{n-2}[\beta^2]$

$= 2023\alpha^n + 2024\beta^n$

$S_{n-1} + S_{n-2} = S_n$

Put $n = 12$

$S_{11} + S_{10} = S_{12}$

64.

$2^m - 2^n = 56$

$2^n(2^{m-n} - 1) = 2^3 \times 7$

$2^n = 2^3$ and $2^{m-n} - 1 = 7$

$\Rightarrow n = 3$ and $2^{m-n} = 8$

$\Rightarrow n = 3$ and $m - n = 3$

$\Rightarrow n = 3$ and $m = 6$

P(6,3) and Q(–2, –3)

$PQ = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$

Hence option (1) is correct

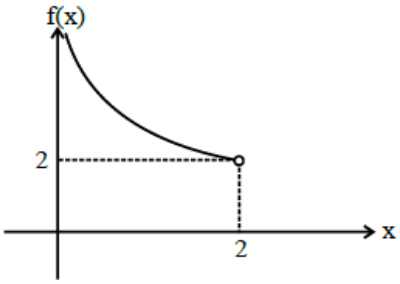
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65.

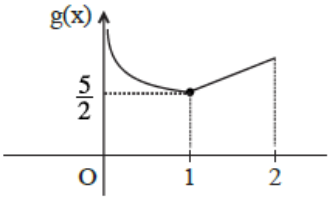
$f:(0,2)\to R;\ f(x)=\frac{x}{2}+\frac{2}{x}$

$f'(x)=\frac{1}{2}-\frac{2}{x^2}$

$\therefore f(x)$ is decreasing in domain.



$$g(x)=\begin{cases}\frac{x}{2}+\frac{2}{x} & 0<x\le 1\\ \frac{3}{2}+x & 1<x<2\end{cases}$$



66.

$f(x)=\frac{2x+3}{2x+1};x\neq-\frac{1}{2}$

$g(x)=\frac{|x|+1}{2x+5},x\neq-\frac{5}{2}$

Domain of f(g(x))

$f(g(x))=\frac{2g(x)+3}{2g(x)+1}$

$x\neq-\frac{5}{2}$ and $\frac{|x|+1}{2x+5}\neq-\frac{1}{2}$

$x\in R-\left\{-\frac{5}{2}\right\}$ and $x\in R$

$\therefore$ Domain will be $R-\left\{-\frac{5}{2}\right\}$

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67.

$$\alpha=\frac{(4!)!}{(4!)^{3!}},\beta=\frac{(5!)!}{(5!)^{4!}}$$

$$\alpha=\frac{(24)!}{(4!)^{6}},\beta=\frac{(120)!}{(5!)^{24}}$$

Let 24 distinct objects are divided into 6 groups of 4 objects in each group.

No. of ways of formation of group $=\frac{24!}{(4!)^6.6!}\in N$

Similarly,
Let 120 distinct objects are divided into 24 groups of 5 objects in each group.
No. of ways of formation of groups

$$=\frac{(120)!}{(5!)^{24}.24!}\in N$$

68.

$$I=\int\frac{x^8-x^2}{(x^{12}+3x^6+1)\tan^{-1}\left(x^3+\frac{1}{x^3}\right)}dx$$

Let $\tan^{-1}\left(x^3+\frac{1}{x^3}\right)=t$

$$\Rightarrow\frac{1}{1+\left(x^3+\frac{1}{x^3}\right)^2}.\left(3x^2-\frac{3}{x^4}\right)dx=dt$$

$$\Rightarrow\frac{x^6}{x^{12}+3x^6+1}.\frac{3x^6-3}{x^4}dx=dt$$

$$I=\frac{1}{3}\int\frac{dt}{t}=\frac{1}{3}\ln|t|+C$$

$$I=\frac{1}{3}\ln\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|+C$$

$$I=\ln\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|^{1/3}+C$$

Hence option (1) is correct

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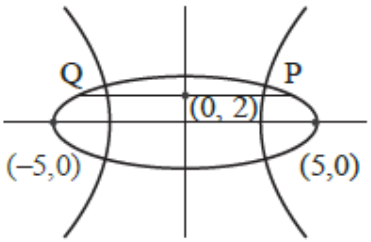
69. $H: \frac{x^2}{16} - \frac{y^2}{9} = 1 \qquad e_1 = \frac{5}{4}$

$\therefore e_1 e_2 = 1 \Rightarrow e_2 = \frac{4}{5}$

Also, ellipse is passing through $(\pm 5, 0)$

$\therefore\ a = 5$ and $b = 3$

$E: \frac{x^2}{25} + \frac{y^2}{9} = 1$



End point of chord are $\left(\pm\frac{5\sqrt{5}}{3}, 2\right)$

$\therefore L_{PQ} = \frac{10\sqrt{5}}{3}$

70. $\frac{{}^6C_4}{{}^{15}C_4} \times \frac{{}^9C_4}{{}^{11}C_4} = \frac{3}{715}$

Hence option (3) is correct.

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71.

$$f(x)=\int_0^x g(t)\ln\left(\frac{1-t}{1+t}\right)dt$$

$$f(-x)=\int_0^{-x} g(t)\ln\left(\frac{1-t}{1+t}\right)dt$$

$$f(-x)=-\int_0^x g(-y)\ln\left(\frac{1+y}{1-y}\right)dy$$

$$=-\int_0^x g(y)\ln\left(\frac{1-y}{1+y}\right)dy \quad (g \text{ is odd})$$

$$f(-x)=-f(x)\Rightarrow f \text{ is also odd}$$

Now,

$$I=\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx \qquad .....(1)$$

$$I=\int_{-\pi/2}^{\pi/2}\left(f(-x)+\frac{x^2e^x\cos x}{1+e^x}\right)dx \qquad .....(2)$$

$$2I=\int_{-\pi/2}^{\pi/2} x^2\cos x\,dx=2\int_0^{\pi/2} x^2\cos x\,dx$$

$$I=\left(x^2\sin x\right)_0^{\pi/2}-\int_0^{\pi/2}2x\sin x dx$$

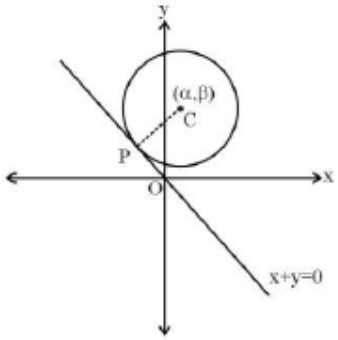
$$=\frac{\pi^2}{4}-2\left(-x\cos x+\int\cos x dx\right)_0^{\pi/2}$$

$$=\frac{\pi^2}{4}-2(0+1)=\frac{\pi^2}{4}-2\Rightarrow\left(\frac{\pi}{2}\right)^2-2$$

$$\therefore \alpha=2$$

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72.



$S:(x-\alpha)^2+(y-\beta)^2=50$

$CP=r$

$$\left|\frac{\alpha+\beta}{\sqrt{2}}\right|=5\sqrt{2}$$

$$\Rightarrow(\alpha+\beta)^2=100$$

73.

$|z-z_0|^2=4$

$\Rightarrow(\alpha-z_0)(\bar{\alpha}-\bar{z}_0)=4$

$\Rightarrow\alpha\bar{\alpha}-\alpha\bar{z}_0-z_0\bar{\alpha}+|z_0|^2=4$

$\Rightarrow|\alpha|^2-\alpha\bar{z}_0-z_0\bar{\alpha}=2$ .....(1)

$|z-z_0|^2=16$

$$\Rightarrow\left(\frac{1}{\bar{\alpha}}-z_0\right)\left(\frac{1}{\alpha}-\bar{z}_0\right)=16$$

$\Rightarrow(1-\bar{\alpha}z_0)(1-\alpha\bar{z}_0)=16|\alpha|^2$

$\Rightarrow1-\bar{\alpha}z_0-\alpha\bar{z}_0+|\alpha|^2|z_0|^2=16|\alpha|^2$

$\Rightarrow1-\bar{\alpha}z_0-\alpha\bar{z}_0=14|\alpha|^2$ .....(2)

From (1) and (2)

$\Rightarrow5|\alpha|^2=1$

$\Rightarrow100|\alpha|^2=20$

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74.

$|A - xI| = 0$

Roots are -1 and 3

Sum of roots $= tr(A) = 2$

Product of roots $= |A| = -3$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

We have $a + d = 2$

$ad - bc = -3$

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$$

We need $a^2 + bc + bc + d^2$

$= a^2 + 2bc + d^2$

$= (a + d)^2 - 2ad + 2bc$

$= 4 - 2(ad - bc)$

$= 4 - 2(-3)$

$= 4 + 6$

$= 10$

75.

$$\frac{dy}{dx} = \frac{x + y - 2}{x - y}$$

$x = X + h, y = Y + k$

$$\frac{dY}{dX} = \frac{X + Y}{X - Y}$$

$$\left.\begin{matrix} h + k - 2 = 0 \\ h - k = 0 \end{matrix}\right\} h = k = 1$$

$Y = vX$

$$v + \frac{dv}{dX} = \frac{1 + v}{1 - v} \Rightarrow X - \frac{dv}{dX} = \frac{1 + v^2}{1 - v}$$

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$$\frac{1-v}{1+v^2}dv=\frac{dX}{X}$$

$$\tan^{-1}v-\frac{1}{2}\ln(1+v^2)=\ln|X|+C$$

As curve is passing through (2, 1)

$$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ln|x-1|$$

$$\therefore \alpha=1 \text{ and } \beta=2$$

$$\Rightarrow 5\beta+\alpha=11$$

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